SOS
BT Waypoint
Overhauled real-time routing for navigation.

Brief Overview

“If I’m on a long walk home past midnight, what’s the safest route, are there safe places? Can I trust the route all-throughout?”

Navigation & Safety - Reality

- Measured **nighttime safety** in Belfast is **inconsistent!**
  - Above **33K** crimes recorded in **2023-24**.
  - **62%** report **anti-social behavior** as an issue.
  - **Confidence** drops in **town-centers** at night **(91% -> 79%)**.
- For data, **the following** must be kept in mind:
  - The risk **always changes. Unpredictable or not**, we must ensure the **correct decisions** are made.
  - Current solutions rely on **historical data** (e.g. NISRA crime datasets) without a good enough system to adapt to **time-of-day risk**. Datasets are published as **aggregated reports (monthly/annual).**

Navigation & Safety – The Gap

- “Safe locations” exist, but a reliable routing system is not built around them:
  - WalkSafe for instance, has a system with labelled safe spaces or zones.
  - Police crime maps plainly use **recorded data** to create “hotspots”.

  Most systems **ignore or poorly consider open venues, lighting/CCTV, live incidents and time of day.**

  There exists a decision **transparency gap**; apps give you a “safe route” however there’s **no explanation** and no **disclosure of factors involved.**

Problem Summary

- Most data-driven safe navigation tools have lacking local, time-dependent routing. They use **static or loosely updated** data.
- Little to no integration of algorithmic decisions over physical location safety and decision into routing logic.
- It is crucial to develop a reliable means of meaningful adjustment which is **time sensitive**, and hence well-suited to real-time commutes of **any length**.

How to get to a Solution

- PSNI, NISRA, etc show data crime varies over time, so we **strictly incorporate time into routing and re-routing**, to ensure routes update **in as little as 10 minutes** (due to **local news,** etc).
- We model safety **as a combination from many more factors** including but not limited to incident data, nearby activity and environment **(from 0-100)**.
- Every route includes a safety score **reason for users to see**.
- With over 44,000 behavioural incidents, the route can respond instead of being fixed (unsafe).

Methodology

Roads on OpenStreetMap (with manual pathfinding) on an A* algorithm are used.
Streets are segmented into a set of points to form an open polygon.

The center point (representative of the particular street) has a 200m geographical radius for searching the police database for any recent criminal activity for the last 45 days.

All criminal activity is also weighted, for example: knife crimes and sexual offences are ranked higher over things like shoplifting incidents. This results in a score out of 100
The closer the crime, the higher the score, and routing responds.

Methodology

The PSNI database API with live data is used to only query the past day for any updates and for any updates from further back are sourced from OpenNI.

The real-time API needs a request a network call every time, running queries for each point will result in 100,000s of queries. So, every 10 minutes, points are checked before an update, for near-real-time functionality (handled by a background worker).

Data on street light location and status from openNI are also used for the overall scoring algorithm.

Methodology

Belfast local news is also fetched via an API. It’s all news and headlines formatted to serve to catch an audience. So use LangChain (which calls featherless API used to use DeepSeek model).

LangChain checks if news is relevant, and what kind of crime it is, where it is and allows us to use news. This implemented as part of the background worker system.

LangChain is used every 10-minutes, which is according to spec. Data fetched is then stored in the database for scoring.

Methodology

Along the street, the system figures out which sanctuaries are currently OPEN, e.g. way more sanctuaries open in your area open in 2 PM vs 2 AM.
A bonus system is applied to the overall safety score obtained, influenced by places which are good for certain times of the day (time of day is factored in) such as public parks.

Street XYZ, per 100m, let’s say 4 crimes happen per week:

The score of 0-100 is logarithmically scaled according to local past trends, incidents along with and the Belfast average score.

Methodology - Py Iceberg

Overpass (tool which uses OpenStreet data) which is used to form a polygon from raw street points.
The tool used to get the sanctuaries, pharmacies, etc is good for location but not for opening and closing time, and also number of places were small. This is where four square is used, which has data points for 100 mil+ places. Polars is used to fetch necessary data from it into memory.
Some the data is more than 5 or even 10 years old, some don’t provide opening hours, some places closed down, so we had to clean data based on whether it was last updated. This is correlated with the overpass data.

Methodology

The segments of each street were joined using a hybrid join data from overpass is only in small chunks, we have to figure out the start and end points to join them smartly to form a full street. After making a full street, small chunks are still used for scoring, presenting on frontend we average the segment scores but for the backend itself we still use chunks for accuracy.

Both kinds of data discussed above are available on the frontend however, we used averaged scores over the full street in an attempt to prevent information overload.

Methodology

We have to find a way to map the sanctuaries and the paths. Using polygon points to snap them together based on their geolocation.

We now have to work on the part where it finds a path from point A to point B using a graph system, Network X (converts data points into a graph problem for traversal).

We find out the nearest neighbouring segments/nodes, once we arrive at an intersection, the A* algorithm uses the safety score as a heuristic to select paths based on whichever is safer. If it's highly unsafe to go down a path, even if its shorter, the longer path will be preferred for a safer journey overall.

Methodology

Even though at every point we were making a decision even when it wasn’t needed.
On a street with say... 2 intersections
on Network X, an optimisation that eliminates redundant nodes between intersections helps speed up the algorithm and make it more reliable.

For the scoring algorithm, we consider user feedback.